

NEURAL BELLMAN OPERATORS

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We develop neural Bellman operators with stopping-accessibility-aware verification and explicit policy-improvement error budgets. In a stopped preference-adjustment economy, we prove that continuation value has a strict jump at an inward wealth boundary, and construct a neural policy and witness that eliminate the trace budget without changing the original control problem. A quantitative certified policy-iteration recurrence separates optimization defects from discretization error. Twenty full-horizon neural checkpoints and six classical state-space candidates are verified in the original economy; their residual bounds remain above the declared accuracy target. In a coupled inventory economy with up to 128 states, a quadratic performance-difference identity gives state-global certificates for all retained neural gain functions, with every final checkpoint below 0.000095606 per coordinate. Separate original-economy comparisons include a full eighteen-node MPFR arithmetic audit and a freshly generated price continuum with regret below 0.009981645276. A precisely defined consumption-top-up bound translates the latter into a scale-normalized compensation measure. The complete numerical record preserves the distinction between a valid certificate, a convergent approximation criterion, and demonstrated policy accuracy.

KEYWORDS: Stochastic control, Neural Bellman operators, Preference adjustment, Validated numerics, Policy verification, Computational economics.

1. INTRODUCTION

A neural solution of a dynamic economic model should be judged by the policy it delivers and the accuracy that can be established for that policy. Average residuals are not a substitute for such an assessment. Stopping boundaries make the distinction particularly important: a boundary observation may describe immediate liquidation without being the limit of continuation value from the interior. A numerical approximation that ignores this distinction can be inconsistent before the choice of optimizer or network width is considered.

This paper develops neural Bellman operators with explicit policy evaluation, improvement, and verification stages. The economic objective is an accurately certified state-contingent policy for the original continuous-time decision problem. We retain the full control opportunity and the original stopping contract when evaluating a candidate. The verification stage is independent of the optimizer’s stopping rule and returns an a posteriori policy-loss bound. We also give a quantitative policy-iteration recurrence whose hypotheses are certified evaluation and improvement defects, rather than sampled training losses.

Our first construction concerns a stopped consumption–investment economy with costly preference adjustment. We prove a strictly positive jump, exceeding 6.37 utility units, between continuation value near its upper wealth boundary and the liquidation payoff at that boundary. Thus a continuous critic constrained to equal the liquidation payoff on every face cannot acquire a vanishing uniform positive Bellman residual. This is a property of the unchanged economy, not a reason to alter its settlement. A signed, accessibility-weighted comparison resolves the inconsistency: the candidate policy cannot reach the upper wealth face, while its witness dominates settlement there and matches settlement on the reachable faces. These properties eliminate the entire trace contribution to its policy certificate without eliminating any action from the optimal upper comparison.

The associated full-horizon neural experiment retains ten seeds, two widths, and both prescribed checkpoints. Each stored policy and critic is enclosed over the complete original state–time cylinder. The boundary construction succeeds exactly, but the current residual bounds remain large: the twenty time-zero policy-loss bounds range from 28.547311 to 102.355570. None meets 10^{-2} . Six newly generated state-space candidates, from controlled Markov-chain and semi-Lagrangian schemes, are evaluated under the same continuous-time contract and common witness. They also do not meet this target. These calculations distinguish an effective correction to the approximation space from a claim that the current nonlinear training procedure is already competitive at high verified accuracy.

Our second application provides an accurate neural dynamic-control calculation in a coupled inventory economy with up to 128 state variables. Neural time-dependent gains are trained against differential Riccati residuals, without optimal-solution labels. A quadratic performance-difference identity and a Lyapunov moment bound turn a complete time enclosure into a certificate on the full state space. Every final checkpoint has per-coordinate loss bounded by 0.000095606 for initial mean squared inventory at most one. The method exploits two invariant modes; a structure-aware classical method can exploit exactly the same structure. The result establishes a fully certified high-dimensional neural policy, not a generic dimension-independent nonlinear solver or a neural advantage over scalar Riccati integration.

Sharp economic comparisons in the original nonlinear economy remain separate from these neural policies. A complete MPFR-directed arithmetic recalculation of all eighteen historical policy/dual nodes yields a price-uniform bound of 0.009969007708. A fresh model-to-library experiment, starting only from model constants and deterministic optimization initializations, generates 20 nodes and a bound of 0.009981645276 over all adjustment prices in $[.5, 8]$. The fresh experiment includes policy construction, dual fitting, adaptive node selection, original stopping evaluation, and exact rational price-envelope construction. We interpret its loss through a precisely specified, externally financed consumption-top-up experiment rather than calling an absolute utility tolerance an economic normalization.

These results have different domains and numerical roles. The original neural and classical feedback certificates cover the full nonlinear state–time domain but are not yet sharp. The time-control libraries give sharp original-economy comparisons at specified initial states and prices. The inventory construction gives a sharp state-global neural policy certificate in a different, structured economy. None of these statements substitutes one numerical object for another. The previous coefficient-transport, state–price continuation, adverse seed comparisons, external holdouts, and frozen-jet action-oracle results remain in the paper and supplement as identified historical evidence.

1.1. *Relation to the literature*

Stopped verification, contraction arguments, and the policy-improvement principle are classical. Judd (1998) develops numerical dynamic programming and economically meaningful approximation assessment; Kushner and Dupuis (2001) studies controlled Markov-chain approximation, and Barles and Souganidis (1991) gives a fundamental monotone-approximation convergence framework. Adaptive sparse grids are an important computational-economics alternative (Brumm and Scheidegger, 2017). Information and control relaxations underpin a substantial literature on dual upper bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural approaches include Han and E (2016), Han, Jentzen, and E (2018), and the retained external comparison uses the implementation of Cai, Fang, and Zhou (2025).

The methodological content here is the identification and correction of a stopping-trace inconsistency in the neural approximation space, the execution of signed policy-specific boundary verification, the separation of certified policy-iteration defects from discretization errors,

and a structure-exploiting state-global neural certificate. The affine-preference dual and joint state-price construction are specialized economic comparison tools. MPFR supplies established directed arithmetic (Fousse et al., 2007); it does not independently prove the shared analytic quadrature or stopping lemmas. We make these dependencies explicit so that reproducibility, arithmetic independence, convergence, and demonstrated policy accuracy are not conflated.

2. THE STOPPED PREFERENCE-ADJUSTMENT ECONOMY

Time is continuous, with horizon $T = 1$. Preferences u and wealth x evolve as

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. The state domain is

$$D = (1.2, 2.8) \times (.5, 2), \quad A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

An admissible control $a = (c, \theta, p)$ is progressively measurable, takes values in A , and is used up to the first exit from D or time one, whichever occurs first. The controls in the original optimization are not restricted to deterministic time functions or to neural networks.

For adjustment price $k \in [.5, 8]$, the running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (4)$$

Let $\tau = 1 \wedge \inf\{s \geq t : (u_s, x_s) \notin D\}$ and $\rho = .04$. The criterion is

$$J_k(t, z; a) = \mathbb{E}_{t,z} \left[\int_t^\tau e^{-\rho(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-\rho(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (5)$$

Here $V_k = \sup_{a \in \mathcal{A}} J_k(\cdot; a)$. The liquidation fee makes a stopping-trace error economically meaningful even when the training distribution assigns little mass near an exit. It is therefore included explicitly in verification rather than approximated by an average boundary loss.

For a smooth test function v , define the discounted residual

$$\begin{aligned} \mathcal{R}_a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\} v_x + .00125 v_{uu} + .02p^2 x^2 v_{xx} - .0025 p x v_{ux} \\ & + \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 - .04v. \end{aligned} \quad (6)$$

All constants in the certificate denote the displayed exact decimals. Training uses floating-point approximations of them, but the certification program encloses the exact constants and treats stored binary64 network weights as exact real numbers.

The central initial condition is $(t, u, x) = (0, 2, 1.25)$. We also study the nondegenerate rectangle

$$K = [1.98, 2.02] \times [1.24, 1.26]. \quad (7)$$

The claim on K is uniform over its interior, not a statement about 72 isolated initial conditions. Its proof is given in Section 7, and its tolerance differs from the sharper central-state tolerance.

3. CONSTRUCTIVE NEURAL VERIFICATION

We first state the verification result for a class of stopped controlled diffusions. Let Z take values in a bounded domain, with drift b_a , covariance matrix Σ_a , running reward f_a , constant discount $\rho \geq 0$, and settlement g on the parabolic boundary. Write

$$\mathcal{R}_a v = v_t + b_a \cdot Dv + \frac{1}{2} \Sigma_a : D^2 v + f_a - \rho v.$$

The assumptions below concern the stochastic model and the computed witness. They do not assume regularity of the unknown optimal value.

ASSUMPTION 1: *For every admissible control, the stopped diffusion and its payoff are well defined. The controlled coefficients and payoffs are bounded on the stopped domain. The candidate feedback π is admissible. The witness v is continuously differentiable in time and twice continuously differentiable in state on a neighborhood of the closed domain, and its derivatives are bounded there. Discounted Itô's formula and optional stopping hold for these bounded stopped quantities.*

The compact economy in Section 2 with the stored smooth, bounded neural feedback satisfies these requirements. Boundedness also permits localization of the stochastic integral before taking expectations.

THEOREM 1—A complete residual and improvement certificate: *Under Assumption 1, suppose verified nonnegative functions $e(s), q(s)$ and a number $b \geq 0$ satisfy, at every state and time,*

$$|\mathcal{R}_\pi v| \leq e(s), \quad 0 \leq \sup_{a \in A} \mathcal{R}_a v - \mathcal{R}_\pi v \leq q(s), \quad |v - g| \leq b \text{ on the stopping boundary.} \quad (8)$$

Then, for every initial state z and time t ,

$$0 \leq V(t, z) - J(t, z; \pi) \leq 2b + \int_t^T e^{-\rho(s-t)} \{2e(s) + q(s)\} ds. \quad (9)$$

The same conclusion applies to any admissible implementation whose action lies in a verified output-error neighborhood of the stored feedback, provided that the residual and improvement inequalities have been enclosed over that neighborhood.

The theorem is deliberately an a posteriori statement. It is useful because every item in (8) has a finite implementation, not because its stochastic comparison identity is new. A training history can propose the witness and policy, but cannot replace any of the inequalities. The proof in Appendix A separates the policy-evaluation and upper-comparison errors; each stopping-trace discrepancy is paid once.

3.1. A finite implementation

Fix a rational rectangular cover of the entire state–time domain, including every stopping face. On each cell, interval forward differentiation gives the value, first derivatives, and relevant second derivatives of the stored network. An action oracle bounds the maximum over the continuous action set. Interval evaluation of the deployed policy gives its residual and the difference from this global maximum. Taking the largest bounds in each time slab produces e and q . Facewise

interval evaluation produces b . Finally, integration of these piecewise-constant bounds yields (9).

A cell is accepted only if all operations have finite, valid enclosures and every potential action branch is covered. No failed or inconclusive cell can be removed from the domain. Refinement replaces a cell by a complete cover of that cell, while keeping previous results in the execution record. The method either returns the certified inequality for the specified stored object or records that a requested accuracy has not been established at the available resolution. This is a stopping criterion for an accuracy claim, not a criterion based on validation loss.

In the economy, the action maximization separates into three scalar problems conditional on the interval-enclosed derivatives. Consumption is strictly concave, and its maximizer is

$$c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u}) \quad \text{when } v_x > 0, \quad c^* = .8 \quad \text{when } v_x \leq 0.$$

The preference action is $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. The portfolio objective is

$$h_1 p + \frac{1}{2} h_2 p^2, \quad h_1 = .06 x v_x - .0025 x v_{ux}, \quad h_2 = .04 x^2 v_{xx}.$$

A negative curvature gives a clipped vertex; nonnegative curvature gives an endpoint maximum. When an interval does not determine the sign, the calculation encloses the whole feasible portfolio interval. Thus uncertainty enlarges the bound instead of excluding an action. These formulas preserve the original continuous control opportunities.

3.2. Transport across model coefficients

The affine dependence on k used later is not a general argument for perturbing dynamics. For the latter we use residual transport. Consider another model with the same state domain, action set, and settlement, but possibly different coefficients and discount. Let its discount be at least $\rho \geq 0$, and let $\tilde{\mathcal{R}}_a$ denote its residual.

PROPOSITION 1—Coefficient transport: *Suppose the original checks (8) hold, and interval coefficient bounds verify*

$$\sup_{a \in A} |\tilde{\mathcal{R}}_a v - \mathcal{R}_a v| \leq d(s) \tag{10}$$

throughout the common domain. If Assumption 1 also holds for the new model, its policy regret is at most

$$2b + \int_t^T e^{-\underline{\rho}(s-t)} \{2e(s) + q(s) + 2d(s)\} ds. \tag{11}$$

If the settlement changes by at most b_{par} , replace b by $b + b_{\text{par}}$.

For example, the modulus in (10) can be evaluated directly from

$$|(\tilde{b}_a - b_a) \cdot Dv| + \frac{1}{2} |(\tilde{\Sigma}_a - \Sigma_a) : D^2 v| + |\tilde{f}_a - f_a| + |\tilde{\rho} - \rho| |v|.$$

It includes the change in the covariance matrix rather than treating volatility perturbations as drift changes. The proposition does not couple paths or equate stopping times across economies. Each model is verified against its own stopped dynamics on the common domain. The executed coefficient box is reported in Section 11.

4. STOPPING ACCESSIBILITY AND THE NEURAL APPROXIMATION SPACE

A stopping contract need not be the limit of continuation value at every geometrical boundary point. This distinction matters for a degenerate controlled diffusion: an investor can eliminate portfolio risk and move away from a wealth boundary. It therefore cannot be decided by assigning a larger penalty to every boundary observation in a neural loss.

PROPOSITION 2—Trace consistency in the unchanged economy: *For the economy in Section 2, at $k = 2$,*

$$\liminf_{x \uparrow 2} V_2(0, 2, x) - g(0, 2, 2) \geq 6.45 - .1 \log 2 - 25.1e^{-8} > 6.37. \quad (12)$$

Consequently, a sequence of witnesses continuous on the closed cylinder, each equal to g on its entire stopping boundary, cannot have its uniform positive Bellman residual converge to zero.

The proposition concerns the consistency of an approximation space, not the existence of a solution to the economic decision problem. The original utility, adapted controls, horizon, and liquidation fee are unchanged. Its implication is constructive: enforce the settlement where the proposed policy can stop, and use the correct one-sided comparison where it cannot. In particular, imposing the upper-wealth settlement as a hard equality would exclude the relevant continuation limit. The proof uses the elementary feasible policy $(c, \theta, p) = (.8, 0, 0)$ and does not use an estimated optimal value.

4.1. A signed, facewise comparison

Partition the stopping boundary into measurable faces F_i , assigning intersections to one face. Let

$$\omega_i^*(t, z) \geq \sup_a \mathbb{E}_{t,z}[e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)], \quad \omega_i^\pi(t, z) \geq \mathbb{E}_{t,z}^\pi[e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)].$$

These are upper bounds on discounted hitting probabilities, not probabilities estimated by simulation. A nonnegative smooth function whose discounted generator is nonpositive and whose stopping trace dominates the face indicator supplies such a bound by optional stopping. A separate invariance argument can give the exact value zero for an inaccessible face.

THEOREM 2—Accessibility-weighted neural verification: *Under Assumption 1, suppose complete enclosures establish*

$$\sup_a \mathcal{R}_a v \leq e_+(s), \quad -\mathcal{R}_\pi v \leq e_-(s), \quad e_+, e_- \geq 0,$$

and $(g - v)_+ \leq \beta_i^+$, $(v - g)_+ \leq \beta_i^-$ on F_i . For an interior initial state,

$$\begin{aligned} 0 \leq V(t, z) - J(t, z; \pi) &\leq \int_t^T e^{-\rho(s-t)} \{e_+(s) + e_-(s)\} ds \\ &\quad + \sum_i \beta_i^+ \omega_i^*(t, z) + \sum_i \beta_i^- \omega_i^\pi(t, z). \end{aligned} \quad (13)$$

The upper comparison retains the entire original action set. Inaccessibility for the candidate policy does not remove an action from that comparison.

This theorem retains the stopped verification identity underlying Theorem 1. The new distinction is which sign of a trace discrepancy must be paid, and under whose stopping distribution it is paid. Replacing both signs by a common supremum discards precisely the boundary information identified by Proposition 2. The inequality is still an a posteriori certificate; convergence of a candidate-generation procedure is a separate statement.

4.2. An executable accessibility-preserving architecture

Let N_c, N_θ, N_p, N_v be smooth neural outputs. We use

$$c = .425 + .375 \tanh N_c, \quad \theta = .2 \tanh N_\theta, \quad (14)$$

$$p = \frac{2-x}{1.5} \{ .15 + .65 \tanh N_p \},$$

$$v = G + (1-t) \{ -8 + B(u, x) \text{softplus}(N_v) \}, \quad (15)$$

$$B(u, x) = (1 - e^{-12(u-1.2)})(1 - e^{-12(2.8-u)})(1 - e^{-4(x-.5)}).$$

Here $\text{softplus}(y) = \log(1 + e^y)$. Consumption and preference controls obey the original bounds. The portfolio head also obeys the original bounds throughout the state domain. Its boundary factor is part of the policy, not a replacement of the economy's portfolio opportunity.

PROPOSITION 3—Exact elimination of the trace budget: *For an interior initial state, the policy in (14) cannot hit $x = 2$ before another stopping event. The witness in (15) equals g on the two preference faces, the lower wealth face, and the terminal face, and is at least g on the upper wealth face. Hence every trace contribution in (13) is zero.*

The statement remains true for bounded admissible perturbations of the action heads, clipped to their stated intervals *before* multiplication of the portfolio head by $(2-x)/1.5$. An unrestricted additive error in the final portfolio output is different: it can destroy inaccessibility and is not covered by this claim. The replication records this implementation contract explicitly. The certificate describes a real mathematical feedback with stored dyadic weights; it is not a claim about an undocumented floating-point activation routine or an SDE simulator.

The architecture does not by itself make the residual small. Full-domain derivative enclosures must still validate both terms in (13). Reporting a zero trace budget without the remaining residual budget would therefore be incomplete. Nor do we force the numerical continuation witness to equal the known immediate-settlement value at an initially stopped upper-boundary state: at that state every policy has value g and regret zero directly.

5. CERTIFIED NEURAL POLICY ITERATION

It is useful to distinguish three sources of approximation: policy evaluation, policy improvement, and approximation of a continuous-time transition problem. We state a finite Bellman result that makes the first two requirements quantitative. It does not assume that a particular stochastic-gradient optimizer attains them.

Consider a finite discounted Markov problem with rewards r_a , stochastic or substochastic transition matrices P_a , and contraction modulus $q \in (0, 1)$. Absorbing settlement is incorporated in the reward. Write $T_a v = r_a + q P_a v$ and $T v = \max_a T_a v$. A neural evaluation v_n of policy π_n and an improved policy π_{n+1} are accepted with recorded defects

$$\|v_n - T_{\pi_n} v_n\|_\infty \leq \epsilon_n, \quad 0 \leq T v_n - T_{\pi_{n+1}} v_n \leq \eta_n \mathbf{1}. \quad (16)$$

On a continuous state space the same inequalities require a complete enclosure, not a finite collection of collocation residuals. On a finite state space they can be checked by enumeration. They give a stopping and refinement criterion for a learned representation without identifying its training loss with its Bellman defect.

THEOREM 3—A quantitative certified policy-iteration recurrence: *Let $E_n = \|V^* - V^{\pi_n}\|_\infty$ for the finite Bellman problem. If (16) holds, then*

$$E_{n+1} \leq qE_n + \frac{2q\epsilon_n}{(1-q)^2} + \frac{\eta_n}{1-q}. \quad (17)$$

Thus geometrically decreasing certified defects imply convergence of policy values, with the explicit convolution bound obtained by iterating (17). If both defects are bounded by Cr^n , $r \in (0, 1)$, the bound decreases at rate $O(\max\{q, r\}^n)$ when $q \neq r$, and $O(nq^n)$ when $q = r$.

The contraction and policy-improvement ingredients are classical. The operational content is the use of independently checked evaluation and continuous-action defects as acceptance criteria. A neural optimizer that does not achieve those defects has not met the hypothesis. In particular, the full-horizon experiments below are not assigned a convergence guarantee by virtue of being called policy iteration.

A time or state discretization requires an additional comparison. If a numerical optimal value differs from continuous-time value by at most ζ_h , and the numerical value of the transferred policy differs from its continuous-time payoff by at most ξ_h , its continuous-time loss is at most $E_n + \zeta_h + \xi_h$. Neither term is assumed zero. An alternative is to bypass both transfers and certify the actual continuous-time feedback with Theorem 2; that is the procedure used for the classical candidates below. At an inward stopping face, imposing an inconsistent Dirichlet approximation does not supply the required ζ_h . Proposition 2 explains why this qualification is material rather than technical.

For a fixed positive discount rate, $q = e^{-\rho h}$ approaches one as h decreases. The factors in (17) must therefore be included in any accuracy allocation. The theorem is not a dimension-free complexity result. The original nonlinear economy still uses a full rectangular certificate; the structured extension in Section 10 avoids a state tensor cover for a different, explicitly proved reason.

6. A SHARP ECONOMIC COMPARISON CONSTRUCTION

A neural certificate should be compared with independently identified bounds for the same economy. We therefore retain the affine-preference market-deflator construction and reimplement its numerical evaluation. The lower policy and the upper comparison object are identified separately from the neural feedback throughout.

6.1. Feasible policies and the original stopping contract

A library policy consists of 16 exact stored consumption constants $c_j \in [.7, .8]$, preference actions $\theta_j \in [0, .2]$, and $p_j = 0$. These controls are held on their time intervals until the original first exit. Wealth is deterministic and satisfies

$$x_1 = e^{.02} \left(x_0 - \int_0^1 e^{-.02s} c_s ds \right).$$

The program proves $x_1 > .5$ and the complete action bounds before evaluating a payoff. It does not round a budget residual to zero or assume that a nearly feasible policy is feasible.

The previous implementation evaluated a polynomial representation with Gaussian moments. The new evaluator instead applies an interval-certified Gauss rule directly to the original CRRA expression on a covered Gaussian domain. After $t = v^2$, it integrates over v and a standard-normal coordinate z . Rationally isolated Gauss nodes, interval weights, and analytic complex-neighborhood remainder bounds enclose the integrals. The unbounded normal tail and the discrepancy from stopping are accounted for analytically. The use of an auxiliary bounded full-horizon extension in this proof does not change the objective (5): it is connected to that objective by an explicit stopping correction, less than 3.87×10^{-18} on K for these policies.

At $k = 4.25$, the second implementation encloses the policy payoff in

$$[-1.315469674845588, -1.315469674834266], \quad (18)$$

with the stored unrounded interval width approximately 1.13×10^{-11} . This is a policy-payoff width, not an optimality gap. The latter still requires an upper bound on V_k .

6.2. The affine-preference dual

Let $dY_s = .02Y_s ds - .3Y_s dZ_s^x$. Its product with wealth satisfies

$$d(Y_s x_s) = (.04Y_s x_s - Y_s c_s) ds + \text{a local martingale.}$$

The cancellation holds for every original adapted portfolio control. Define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad \beta = \partial_u \Psi.$$

For a deterministic reference $\bar{\theta} \in [0, .2]$, set $m(t) = 2 + \int_0^t \bar{\theta}_s ds$ and let Q be the affine tangent to $\phi(u) = -.02(u-2)^2$ at $m(1)$. On $u \in [1.5, 2.5]$, $y \in [.2, 12]$, the supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (19)$$

It is justified by analytic monotonicity and complete interval covers, not by sampled tangency or presumed global concavity. Its new primitive checks have strictly positive margins; the supplement gives the proof and the recorded numerical margins.

Put $\hat{y} = \text{proj}_{[.2, 12]} y$ and replace the source by the right side of (19) at $\hat{y}$. The resulting full-line auxiliary problem $A_k(t, u, y)$ retains adapted preference actions, quadratic adjustment cost, and terminal payoff Q . Its source and terminal value are affine in the initial preference. The following positive localization potentials reconnect it to the original first-exit problem:

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (20)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (21)$$

For $y_0 \in [1.2, 2.4]$ the resulting upper bound is

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (22)$$

The auxiliary value has a computable upper bound consisting of one-Gaussian source expectations, a terminal tangent, a correlated-shock correction, a reference-action gap, and a

conditional-variance allowance. The latter is necessary because the adapted marginal value of adjustment is random; replacing it by its unconditional mean without a correction would not be valid. The supplement states and proves the complete formula, including the factors .00375 and .09/(24k) in the covariance and variance terms.

In the second implementation, the source uses separate analytic formulas for its capped and interior consumption branches. Interval automatic differentiation supplies Hessian enclosures, and a rational-root Gauss construction encloses the weighted moments. This differs from the inherited manually differentiated source and polynomial moment integrator. The analytic dual theorem is shared; its mathematical validity is not established merely by agreement between programs. Its finite supporting-plane checks and all decisive numerical quantities are recomputed by the new implementation.

7. JOINT INITIAL-STATE AND PRICE CERTIFICATION

The state-domain extension is obtained without assuming a Lipschitz modulus for the unknown optimal value. Two structural properties suffice: a convex upper comparison function and a concave feasible-policy lower function.

7.1. Extending the dual in the initial state

For the fixed dual witness, define

$$b_0 = \int_0^1 e^{-\rho s} \mathbb{E} \beta(m(s), \hat{Y}_s) ds + e^{-\rho} Q'. \quad (23)$$

An interval for b_0 is obtained in the same independent execution as the dual. Since the coefficient of the initial u in the auxiliary criterion is independent of the chosen adapted preference action,

$$A_k(0, u, y_0) = A_k(0, 2, y_0) + (u - 2)b_0.$$

Let U_0 be a certified upper endpoint in (22). It follows that

$$\begin{aligned} V_k(0, u, x) &\leq U_0 + y_0(x - 1.25) + (u - 2)b_0 \\ &\quad + 8e^{-18} \{e^{120(u-2)} + e^{-120(u-2)} - 2\}, \end{aligned} \quad (24)$$

for the verified initial rectangle. Interval multiplication is sign-aware when $u - 2 < 0$. With an interval for b_0 , its upper support function is convex in u . Thus the computable right side of (24) is convex in (u, x) . This is a property of the upper comparison object, not an assertion that V_k itself is convex in wealth.

7.2. A budget-preserving policy map

Write $Q_r = \int_0^1 e^{-.02s} ds$. For each stored policy, define

$$c_j(x) = c_j + \frac{x - 1.25}{Q_r}, \quad \theta_j(x) = \theta_j, \quad p_j(x) = 0. \quad (25)$$

Its terminal wealth is identical to that of the central-state policy. All shifted consumption values remain in $[.7, .8]$ on K ; this is checked for every library member. The policy depends on the

initial state and is thereafter a time control until exit. It is not described as an approximate neural feedback surface.

The utility $f(c, u) = c^{1-u}/(1-u)$ is jointly concave over the region needed by the Gaussian cover. With $r = u - 1$ its Hessian satisfies

$$\begin{aligned} f_{cc} &= -uc^{-u-1}, \\ f_{cu} &= -(\log c)c^{-u}, \\ f_{uu} &= -\frac{c^{-r}}{r^3} \{(r \log c + 1)^2 + 1\}, \\ \det D^2 f &= \frac{c^{-2u}}{r^3} \{r^2(\log c)^2 + 2r(r+1) \log c + 2(r+1)\}. \end{aligned}$$

On $u \in [1.2, 2.8]$, $c \in [.7, .8]$, one has $r(-\log c) < 1$, so the determinant is positive. The covered expected payoff, with the exact budget shift, is therefore concave in the initial state. A single uniform tail and stopping correction turns this covered functional into a lower bound for the original stopped policy throughout K .

PROPOSITION 4—Four vertices and a continuum of prices: *Suppose, at each price node k_i , a convex upper function $U_i(z)$ bounds $V_{k_i}(0, z)$ on a rectangle K . For each policy j , suppose a concave lower function $F_j(z; k)$ bounds its original payoff on K , is affine in k , and has a verified expenditure slope interval. Then corner values alone produce a uniform regret certificate on $K \times [k_1, k_N]$ by the following finite construction. Between adjacent price nodes use the chord of U_i and U_{i+1} . For each policy transfer its four lower corner values affinely in price, using the appropriate expenditure endpoint. At each price select the policy minimizing the largest of its four upper-minus-lower corner gaps. The supremum of the resulting bound is attained at an endpoint or an intersection of the finitely many affine corner-gap functions.*

The optimal value is convex in k because the policy set, dynamics, and stopping contract are common and fixed-policy payoffs have the form $A(\pi) - kB(\pi)$. This justifies the upper price chord, not interpolation of V as though it were affine. The expenditure used for the concave covered lower functional is its full-horizon deterministic expenditure, with the stopping discrepancy already paid uniformly at the largest price. For the separate central-state exact-policy envelope, the program instead bounds the stopped expenditure. Confusing these two slopes would invalidate the transfer; they are separately recomputed and labeled in the evidence.

PROPOSITION 5—Executed state–price certificate: *For the unchanged economy (5), the 18 stored policies with the state map (25) and the price selector in Proposition 4 satisfy*

$$\sup_{(u,x) \in K} \sup_{k \in [.5, .8]} \{V_k(0, u, x) - J_k(0, u, x; \pi_{k,x})\} \leq 0.012095431554. \quad (26)$$

The central-state selector based on the independently evaluated stopped policies satisfies

$$\sup_{k \in [.5, .8]} \{V_k(0, 2, 1.25) - J_k(0, 2, 1.25; \pi_k)\} \leq 0.009969007711. \quad (27)$$

The execution evaluates all 72 policy–state corners, uses all 18 independently verified upper witnesses, and checks 211 exact candidate breakpoints for the joint state–price construction. No

unsampled state requires an empirical smoothness extrapolation. The exact rational worst-case joint bound is

$$\frac{2187960974729572475442514300375}{180891518006739738096917284913152}.$$

The displayed decimal in (26) is rounded upward. The computational record contains the complete selected-policy regions and every corner interval. Appendix B proves the interpolation and finite-breakpoint steps.

8. FULL-HORIZON NEURAL AND CLASSICAL EXPERIMENTS

The original nonlinear economy is unchanged in all experiments in this section. The architecture is (14)–(15). Seeds 16100–16109 use two hidden tanh layers, with width 16 for even seeds and width 32 for odd seeds. Every trajectory retains the 800- and 2,400-update checkpoints. Each actor update follows three critic updates; Adam learning rates are .0002 and .0005, respectively, with batch size 128, double precision, and one Torch thread. A quarter of each sampling mixture targets preference and lower-wealth boundary layers. No optimal-value labels are used. These settings were fixed before the canonical remote execution; no best-seed substitution is made.

For every checkpoint, MPFR interval differentiation encloses the complete $[0, 1] \times [1.2, 2.8] \times [.5, 2]$ cylinder on $4 \times 16 \times 16$ cells. The continuous consumption, adjustment, and portfolio maxima use the original action set. The certificate allows a componentwise head perturbation of 2^{-24} , clipped to the head bounds before the portfolio-distance multiplication. This location of the perturbation matters: an unrestricted additive error in the final portfolio could invalidate the boundary-inaccessibility argument. The proof concerns mathematical real-valued networks with stored dyadic weights under this explicit output contract. It does not certify a particular machine-library tanh implementation or an SDE simulation stack.

TABLE I
FULL-HORIZON NEURAL CHECKPOINTS IN THE ORIGINAL ECONOMY

Seed	Width	800 updates	2,400 updates	Train (s)	Verify (s)
16100	16	30.19468	54.00405	165.70	28.36
16101	32	41.45535	96.87386	180.11	79.46
16102	16	28.84397	53.60464	156.26	29.22
16103	32	42.60784	93.28879	162.35	59.47
16104	16	29.69505	68.44466	138.88	23.06
16105	32	48.37134	100.68512	139.71	65.74
16106	16	30.03684	59.19474	129.16	27.60
16107	32	45.64637	102.35557	143.36	62.72
16108	16	28.54732	49.67500	132.95	25.88
16109	32	44.11682	95.77234	74.49	23.67

Each bound is uniform over the original interior state rectangle at time zero, $k = 2$, and uses 1,024 cells with the original continuous action maximum. The trace contribution is exactly zero. Training time covers both checkpoints in one trajectory; verification time sums their two audits. No checkpoint meets 10^{-2} . Resource measurements are descriptive, not matched-accuracy efficiency comparisons.

Table I retains every prescribed checkpoint. The exact trace budget is zero, but the positive optimal residual and negative policy residual still produce large bounds. An increase in optimization budget does not uniformly tighten these certificates. Since the bounds also include interval dependency, this observation does not by itself prove that the realized policy payoff

deteriorates. The refinement experiment holds the first seed’s final network fixed and doubles the cover in each coordinate; the complete cell records separately retain the signed budget and the more conservative unsigned residual/action budget. The hard-all-face ablation is retained as well. Its finite-resolution certificate is not evidence that the excluded approximation space becomes consistent as residual error tends to zero.

8.1. State-space candidate generators

We implement a controlled Markov-chain approximation with nonnegative transition weights and a semi-Lagrangian weak Euler approximation with correlated shocks. Each uses the same utility, discount, action bounds, state domain, and first-exit settlement. Refinement uses 9, 17, and 25 nodes per state coordinate, 64, 128, and 256 time steps, and progressively richer bounded control-head grids. The Markov-chain implementation uses wide diffusion jumps and a checked nonnegative stay probability; the semi-Lagrangian method uses four symmetric shock directions. Approximate trajectories apply the liquidation contract at their first straight-segment exit. This discretization is a candidate-generation device, not an asserted exact solution of the stopped diffusion.

For verification, each numerical policy is extended by bilinear interpolation in state and piecewise-constant interpolation in time. Its portfolio head is multiplied by $(2 - x)/1.5$, exactly as in the neural candidate class. These are feasible policies in the unchanged economy, although the candidate class is narrower than the original collection of all admissible controls. The upper comparison still uses every original control. We enclose the interpolated policies with the same complete verifier and the fixed seed 16100 final critic. Thus their discrete value outputs are never used as continuous-time upper or lower bounds.

TABLE II
SAME-ECONOMY STATE-SPACE CANDIDATE GENERATION AND VERIFICATION

Method	Nodes/axis	Time steps	Regret bound	Generate (s)	Verify (s)
Markov chain	9	64	48.98350	0.12	3.84
Markov chain	17	128	48.95819	3.41	3.73
Markov chain	25	256	49.45469	42.90	3.74
Semi-Lagrangian	9	64	48.93352	0.08	3.67
Semi-Lagrangian	17	128	49.47658	1.06	3.69
Semi-Lagrangian	25	256	49.69852	25.90	3.74

All six policies are continuous bilinear state interpolants of bounded heads, piecewise constant in time, with the portfolio-distance factor. The common witness is the predeclared seed 16100 final critic, not a fitted value from either baseline. Generation discretizes the model; the separate certificate evaluates the continuous-time policy and unchanged stopping contract. None meets 10^{-2} , so this is not a matched-accuracy ranking.

The common witness makes the comparison interpretable, but it is currently inaccurate. None of these six candidates attains the 10^{-2} target under that certificate. The experiment supplies two distinct state-space constructions, three refinements, and actual continuous-time policy verification; it does not establish a matched- 10^{-2} efficiency frontier or an optimal implementation of either classical method. In particular, a loose common upper witness cannot rank realized policy quality merely by the size of a certified upper bound. Stronger policy-specific evaluations and upper witnesses remain a numerical task, not an omitted term that can be declared negligible.

9. ARITHMETIC INDEPENDENCE AND FRESH PRICE-CONTINUUM CONSTRUCTION

We distinguish two complete calculations. The first re-evaluates all eighteen historical policy/dual nodes with a new arithmetic implementation. Every basic operation and every exponential, logarithm, square root, and tanh endpoint is evaluated using 128-bit MPFR directed rounding, followed by directed conversion to binary64 endpoints. Stored network weights and policy constants are interpreted as exact dyadic inputs, while model decimals are enclosed as exact rationals. Invalid domains and division through zero fail closed. The kernel passes 450 exact-rational regression checks, but these checks are diagnostics rather than a formal proof of the implementation.

The full-node MPFR calculation recomputes stopped policy payoffs, dual source integrals, expenditure intervals, and the final exact rational envelope. It does not read historical certified endpoints as inputs. It does share the R14 analytic decomposition, rational Gauss-node isolation, derivative remainder lemmas, and stopping inequalities. Thus it is an independent established-arithmetic cross-check of the decisive calculation, not an independent discovery of every mathematical premise. Its complete price-uniform bound is 0.009969007708; full lower and upper intervals, not three pointwise comparisons, are retained at every node.

The second calculation starts from the economic model alone. It initializes the price set at $1/2$ and 8, generates a 16-slab feasible time-control policy at each node by deterministic SLSQP initialization, and fits the corresponding dual from deterministic starts. It then selects the worst exact current envelope breakpoint, rounded to a spacing of $1/1024$; a repeated node triggers bisection of the widest price interval. The declared stopping rule is a proved uniform gap below .01 or thirty freshly generated nodes. No inherited policy, dual pilot, or historically selected intermediate node is read. Optimizer attempts, unsuccessful intermediate envelopes, and resource use are retained.

TABLE III
COMPLETE PRICE-CONTINUUM CALCULATIONS ON THE ORIGINAL ECONOMY

Calculation	Nodes	Uniform regret	Wall time (s)	Peak RSS (MiB)
Historical policies; MPFR	18	0.009969007708	1065.89	276.3
Fresh model-to-library	20	0.009981645276	415.79	497.5

Both cover $(t, u, x) = (0, 2, 1.25)$ and every $k \in [.5, 8]$. Historical inputs are frozen policies and dual pilots, not inherited endpoints; MPFR recomputes every node. The fresh calculation starts from model constants and deterministic initializations, with no historical policy or pilot. Its arithmetic is the retained rational–Taylor implementation; the MPFR label does not apply to the fresh library. Peak MPFR RSS is the largest worker, not summed concurrent memory.

The fresh pipeline reaches 0.009981645276 with 20 generated nodes. This result belongs to the time-control/dual method, not to the neural feedback. The fresh calculation uses the retained rational–Taylor interval kernel; the independent MPFR label belongs only to the separately re-evaluated eighteen-node object. Fresh generation closes a reproducibility distinction without removing the dual relaxation’s accuracy floor. The original 16/32/64-slab experiment and its failure to attain .005 are retained below. Neither extra price nodes nor quadrature refinement alone supplies a convergence law to arbitrarily small error.

The canonical replication command executes all new neural trajectories and checks, both state-space solvers, the fresh price library, the historical-node MPFR audit, the structured inventory experiment, and the normalization calculations. Source identities, input hashes, complete cell records, failures, and hardware/software metadata accompany the results. Timings in different

concurrent experiment processes are descriptive; they are not used to infer solver superiority from unequal accuracy or unequal processor contention.

10. A COUPLED HIGH-DIMENSIONAL INVENTORY ECONOMY

To test full dynamic certification beyond two states, consider a quadratic inventory-adjustment economy. With $M = \mathbf{1}\mathbf{1}'/d$,

$$dX_s = (AX_s + a_s)ds + .1dW_s, \quad A = -.2I + .3M, \quad (28)$$

and minimize

$$\mathcal{J}(t, x; a) = \mathbb{E} \left[\int_t^1 \{X'_s(I + .5M)X_s + |a_s|^2\} ds + .5|X_1|^2 \right]. \quad (29)$$

Controls are square-integrable adapted adjustments. Mean inventory affects drift and aggregate holding cost; idiosyncratic inventory also has its own adjustment dynamics and shocks. This is a coupled d -state dynamic problem, not an action maximization at frozen derivatives. It is a structured, uncalibrated application, and the structure is available to classical solvers as well.

The orthogonal and mean modes have $(a_\perp, q_\perp) = (-.2, 1)$ and $(a_m, q_m) = (.1, 1.5)$. Their Riccati gains satisfy

$$(p_i^*)' + 2a_i p_i^* - (p_i^*)^2 + q_i = 0, \quad p_i^*(1) = .5.$$

We train two neural gain outputs $\hat{p}_i(t) = .5 + (1 - t) \text{softplus}(N_i(t))$ against these *differential residuals*, not against exact Riccati solutions. The feedback is

$$\hat{a}(t, x) = -\hat{P}(t)x, \quad \hat{P} = \hat{p}_\perp(I - M) + \hat{p}_m M.$$

The network changes its gain with time and controls all d state coordinates. One learned pair is reused across the six dimensions; this reuse is not counted as six independent training runs.

THEOREM 4—State-global quadratic neural policy certificate: *Suppose interval differentiation on a complete time cover bounds the two neural Riccati residuals by ε_i . Put $\kappa_\perp = 1.4$, $\kappa_m = .8$, and*

$$D = \max_i \varepsilon_i \frac{1 - e^{-\kappa_i}}{\kappa_i}, \quad C = \frac{1 - e^{-.8}}{.8}.$$

For every $d \geq 1$ and every $|x|^2/d \leq 1$,

$$0 \leq \frac{\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x)}{d} \leq D^2 \left\{ C + \frac{.01}{.8} (1 - C) \right\}. \quad (30)$$

The bound concerns the continuous-time controlled diffusion on all of $\mathbb{R}^d$, not a finite collection of initial states.

The proof uses a quadratic performance-difference identity and a closed-loop Lyapunov moment bound. It replaces state covering by algebra valid on the entire state space. The certificate's time-enclosure work depends on network size and the number of time cells, rather than a tensor grid in d . Applying the feedback requires a mean and coordinatewise operations,

hence $O(d)$ work. This advantage comes from the model’s two invariant modes; it is not a generic escape from dimensional dependence for nonlinear Bellman equations.

Ten fixed seeds, two widths, and both predeclared checkpoints are retained. Every final checkpoint has per-coordinate certified loss at most 0.000095606; the complete range and costs appear in Table IV. The certificate scales to total loss by multiplication by d . The supplement also reports the earlier checkpoint and fixed-network time-cover refinements separately. A structure-aware classical calculation solves only two scalar Riccati equations. Its low cost is reported rather than hidden: this application establishes accurate, reproducibly generated neural feedback with full dynamic certification, not superiority over the classical solution of this specially structured problem.

TABLE IV
CERTIFIED NEURAL POLICY LOSS PER INVENTORY COORDINATE

Seed	Width	250 updates	1,000 updates	Train (s)	Verify (s)
16200	8	0.00117290	0.00009561	5.28	1.18
16201	16	0.00045470	0.00007661	7.05	2.85
16202	8	0.00051798	0.00006084	5.40	1.27
16203	16	0.00011929	0.00003574	7.44	3.26
16204	8	0.00179439	0.00008973	5.09	1.09
16205	16	0.00054105	0.00002530	6.28	3.05
16206	8	0.00036825	0.00004136	5.59	1.36
16207	16	0.00023854	0.00009053	6.34	3.34
16208	8	0.00099196	0.00005105	5.88	1.60
16209	16	0.00036174	0.00007427	5.08	2.29

Each checkpoint uses a complete 512-cell time cover and a state-global quadratic proof. Bounds hold for every $|x_0|^2/d \leq 1$. Multiply the reported bound by d for total loss at $d = 4, 8, 16, 32, 64, 128$. One gain pair is reused across dimensions. Timings cover one training trajectory and its two checks, not six duplicated runs.

10.0.0.1. *Historical evidence retained.* The following original-economy experiments, bounds, comparisons, and welfare-resolution calculations are the R14 research record. Their sources and results are unchanged. They are not additional R16 training runs or endpoints of the fresh R16 library.

11. RETAINED R14: EXECUTED NEURAL VERIFICATION ON THE ORIGINAL ECONOMY

The retained R14 neural experiment uses a state-feedback actor and critic with two hidden layers of 48 tanh units. Inputs are $(2t - 1, (u - 2)/.8, (x - 1.25)/.75)$. The actor’s output transformation is

$$c = .425 + .375 \tanh a_1, \quad \theta = .2 \tanh a_2, \quad p = .15 + .65 \tanh a_3.$$

The critic is $v(t, u, x) = G(u, x) + (1 - t)N(t, u, x)$, so its terminal trace is exact. The first-exit trace is not imposed by this architecture and must be checked.

All weights are freshly generated from seed 1401. Each actor-improvement step is preceded by three critic residual updates, with batch size 128. Adam learning rates are .0005 for the critic and .0002 for the actor, and the boundary-loss coefficient is .025. Snapshots at 100, 300, and 800 updates are retained. The experiment uses the 48-by-48 tanh architecture of the retained comparisons, but its continuous economic residual and boundary objective are a new adapter;

TABLE V
WHOLE-DOMAIN NEURAL CERTIFICATES ON THE UNCHANGED ECONOMY, $k = 2$

Updates	Cells	$2b$	$\int e^{-\rho s}(2e + q)ds$	Regret upper	Seconds
100	2048	14.9778	8.8713	23.8491	4.40
300	2048	14.2970	9.0431	23.3400	4.75
800	256	14.2484	25.9010	40.1494	0.42
800	2048	14.1073	16.1408	30.2480	4.41
800	16384	14.0463	9.4470	23.4932	25.09

Upper bounds are rounded upward. Time is the measured verification stage, excluding training. Every row covers the complete domain with zero skipped and zero failed cells. The last three rows refine the cover for the same 800-update snapshot. None attains the .01 target.

TABLE VI
STOPPING-TRACE ERRORS FOR THE FINEST NEURAL COVER

Stopping face	Cells	Trace error upper
$u = 1.2$	512	4.63971373
$u = 2.8$	512	7.02049755
$x = .5$	512	5.17780198
$x = 2$	512	7.02310795

Each entry is a uniform bound over the complete face and time interval. The comparison theorem pays the largest face bound twice. The error is an evaluation of the stored critic against $G - 8(1 - t)$, not a change in that settlement.

it is not the same finite-step training objective as the external benchmark. The optimizer and sampled objective are proposal mechanisms only.

The certification program uses an independent interval implementation for network evaluation and differentiation. It checks all cells of $[0, 1] \times [1.2, 2.8] \times [.5, 2]$, all four first-exit faces, and the terminal face. A componentwise output neighborhood of radius 2^{-24} is included in the action ranges. This proves robustness for any admissible action within that neighborhood of the mathematical real-tanh network. It does not assert an undocumented error bound for a machine-library tanh routine or for a discretized diffusion simulator.

Table V separates the stopping trace from the integrated policy residual and action gap. The finest cover has 16,384 interior cells and 512 cells on each stopping face. It gives the time-zero bound 23.493172463 and an all-initial-times bound below 23.656877772. The terminal trace is zero. The 800-update sampled residual mean square is approximately .0079446, but this does not control either the worst-case residual or the first-exit trace. Moreover, the 800-update snapshot is not uniformly better than the 300-update snapshot at the same coarse cover. Both outcomes remain in the record.

The coefficient-transport calculation uses the same stored snapshot and cover, with

$$r \in [.019, .021], \quad \mu \in [.058, .062], \quad \sigma_u \in [.049, .051], \quad \sigma_x \in [.195, .205],$$

$$\rho \in [.039, .041], \quad k \in [1.9, 2.1],$$

and correlation fixed at $-.25$. Here μ denotes the risky-wealth drift coefficient multiplying p . The action bounds, state domain, and settlement are unchanged. Proposition 1 gives a time-zero regret bound below 23.530407814 uniformly over this model box. This demonstrates an executed transport mechanism beyond affine objective continuation. It does not make the base neural certificate accurate: the same boundary and approximation errors remain visible.

TABLE VII
INDEPENDENT CENTRAL-STATE CERTIFICATES FOR ALL RETAINED PRICES

k	Policy lower	Optimum upper	Regret upper	Policy width
0.5	-1.27266301	-1.26281817	0.009844825	1.13e-11
0.875	-1.27897331	-1.26925551	0.009717794	1.13e-11
1.25	-1.28479324	-1.27510918	0.009684054	1.13e-11
1.625	-1.29013398	-1.28045672	0.009677256	1.13e-11
2	-1.29501788	-1.28533017	0.009687701	1.13e-11
2.5	-1.30083961	-1.29112854	0.009711056	1.13e-11
3	-1.30590538	-1.29616492	0.009740453	1.13e-11
3.5	-1.31024866	-1.30047467	0.009773977	1.13e-11
4	-1.31389776	-1.30408950	0.009808244	1.13e-11
4.25	-1.31546968	-1.30564683	0.009822839	1.13e-11
4.5	-1.31688907	-1.30705332	0.009835742	1.13e-11
5	-1.31935183	-1.30949330	0.009858515	1.13e-11
5.5	-1.32141556	-1.31153758	0.009877966	1.13e-11
6	-1.32317048	-1.31327568	0.009894785	1.13e-11
6.5	-1.32468139	-1.31477191	0.009909467	1.14e-11
7	-1.32599609	-1.31607368	0.009922399	1.14e-11
7.5	-1.32715060	-1.31721673	0.009933858	1.14e-11
8	-1.32817262	-1.31822851	0.009944109	1.14e-11

Every upper endpoint covers all original adapted controls; every lower endpoint belongs to the identified feasible policy. The last column is the width of that policy's payoff enclosure, not its regret. Regret uses unrounded endpoints. All 18 cases are included, and no inherited generation time is counted as zero.

The finest computation records separate cellwise evaluation, action, and transport quantities. A diagnostic comparison of the interval neural derivatives with automatic differentiation checks 140 derivative components at 20 states and 60 actor components, with no discrepancy outside the numerical tolerance. This diagnostic detects implementation disagreement; it is not substituted for the full-cell interval proof.

12. RETAINED R14: INDEPENDENT NUMERICAL EVIDENCE AND ROBUSTNESS

The sharp economic comparison uses a second implementation that imports none of the inherited primal or dual certifiers. Its basic arithmetic encloses exact rational constants and exact stored binary64 inputs. Addition, multiplication, division, and reductions are rounded outward. Exponentials and logarithms are bounded by rational series with analytic remainders; $\tanh$ is obtained from the enclosed exponential. The Gauss nodes are isolated by exact polynomial signs and the weights are enclosed by their defining formula. The supplement states the floating-point model and the analytic remainder bounds, so these assertions are not based solely on rerunning a production function.

The independent primal evaluator uses 262,144 integrand evaluations per 16-interval policy. The fine dual uses 262,144 source boxes per price. Complete supporting-plane covers certify a negative second-derivative upper bound below $-.06717$ on the first preference region and a strict derivative separation on the remaining region. They also certify a source floor above -6.98956 , a source ceiling below zero, and an absolute preference coefficient below $.97119$. These independently computed margins imply the analytic bounds $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ used for stopping continuation. The complete old policy and dual inputs are retained and identified by their content hashes.

TABLE VIII
SEPARATE QUANTITIES IN THE INDEPENDENT $k = 4.25$ CALCULATION

Quantity	Lower	Upper
Source integral	-2.38605836997	-2.38604817428
Covariance	-0.00017879144	-0.00016944433
Reference-control gap	0	0.00003256016
Conditional variance	0.00003697298	0.00003697299
Localization	2.4594E-7	2.4595E-7
Primal payoff	-1.315469674846	-1.315469674834

Intervals are displayed outward. The source-remainder enclosure, Gaussian tail, terminal tangent, reference cost, and arithmetic allowances are present in the machine-readable record. The conditional-variance allowance is positive and is not removed when quadrature is refined.

TABLE IX
CONTINUUM CERTIFICATES UNDER INDEPENDENT VERIFICATION AND PERTURBATION

Certificate	Uniform regret upper
R12 retained implementation, original inputs	0.009994024847
R14 independent implementation, finer cover	0.009969007711
R14 perturbed witnesses and enlarged allowances	0.009970977450
R14 joint state–cost rectangle	0.012095431554

The first three rows refer to the central initial state. The fourth covers the additional initial-state rectangle K and is not compared with the central-state .01 tolerance as though the domains were identical. The perturbation experiment changes proof witnesses and allowances, not the economic primitives.

Exact price postprocessing is independently implemented by enumerating affine intersections with rational arithmetic, rather than importing the original hull routine. Its input expenditures are newly computed from the stored preference actions with an explicit stopping correction. Thus the upstream policy values, upper values, expenditure intervals, and downstream envelope all have a new execution path. Reproducibility of this path remains distinct from independence of its analytic premises.

12.1. Stress tests for the entire library

The robustness experiment changes every dual pilot: y_0 is increased by 10^{-6} , and its reference preference drifts receive alternating perturbations of $\pm 10^{-6}$ with inward projection into the admissible reference interval. Simultaneously, the proof uses the weaker source floor -7.11 , the larger coefficient bound 1.11 , twice the Gaussian-tail allowance, and 1.1 times the conditional-variance allowance. The resulting continuation floor remains strictly above -8 . All 18 upper certificates are recomputed, while the original economy and feasible lower policies remain unchanged.

The unperturbed independent bound leaves approximately 3.10×10^{-5} below .01, compared with approximately 5.98×10^{-6} for the retained R12 bound. The perturbed calculation remains below .01. Neither statement claims a bound below .009. An additional exact sensitivity calculation shows how a common enlargement of every upper and lower endpoint changes the final envelope: enlarging each side by δ increases the guaranteed gap by at most 2δ . It is a sensitivity result conditional on valid node intervals, not a replacement for the independent primitive audit.

A third, materially different decomposition integrates the capped and interior conjugate using closed truncated-lognormal moments. At prices .5, 4.25, and 8, a separate program computes the

TABLE X
FRESH GENERATION AND INDEPENDENT VERIFICATION AS POLICY RICHNESS INCREASES

Slabs	Generation	Dual fit	Primal audit	Dual audit	Regret upper
16	0.235	0.284	0.737	7.408	0.009688058
32	0.364	0.434	0.926	6.815	0.009669548
64	0.847	0.379	1.791	5.748	0.009665074

All stage times are seconds. Each case starts without inherited policy or dual inputs, and all three optimizations report success. End-to-end measured case times are 8.67, 8.54, and 8.77 seconds. The experiment changes policy richness, unlike a price-node refinement experiment. All cases pass .01; none passes .005, .0025, or .001.

source integral, initial affine coefficient, and covariance term at 40 and 70 decimal digits. Every computed quantity lies inside the independently certified interval. The agreement provides a useful cross-check of algebra and decomposition; these point-valued high-precision quadratures are not themselves rigorous enclosures, and their convergence is not used as a proof of the interval certificate.

13. RETAINED R14: ACCURACY, COMPUTATIONAL WORK, AND COMPARATORS

A certification time measured from frozen proof objects is not the cost of generating those objects. The 18-node independent audit takes approximately 103.08 seconds of measured wall time in the retained run, but starts from existing policies and dual pilots. The 72-corner state audit takes approximately 29.71 seconds. These are verification costs. The generation costs of inherited and interrupted historical attempts are not reconstructed from their absence and are not assigned zero.

The fresh neural run records all 800 actor updates and 2,400 critic updates, the fixed seed, every retained checkpoint, and resource usage. Its recorded execution after program entry is approximately 13.08 seconds, with interpreter and import time separately recorded. It uses one Torch CPU thread and no GPU. The execution environment reports an AMD EPYC 9V74 processor and five exposed logical CPUs; this identifies the environment, not exclusive ownership of a physical machine. Stage times recorded during overlapping local experiments are descriptive clocks, not controlled hardware-efficiency comparisons.

13.1. *A newly generated policy-richness frontier*

The same original economy at $k = 2$ is also solved within deterministic time-control classes of 16, 32, and 64 intervals. Each problem starts from a fixed feasible constant-consumption construction and preference action .05. Constrained sequential quadratic optimization uses analytic gradients to propose a policy. A separately initialized scalar-deflator and reference-drift fit proposes its upper witness. The fresh policy is then checked by the independent stopped evaluator and the fresh upper witness by the independent dual. The fitting objective is not counted as a certificate.

The frontier is informative because its approximation object is held distinct from its upper relaxation. Policy refinement improves the lower bound, but the independently fitted affine upper bound is nearly unchanged. A fixed specialized relaxation can therefore limit the attainable regret even when quadrature error is very small. This calculation is not evidence of convergence to arbitrary tolerance, and it does not establish convergence of the neural optimizer.

The classical comparator is a freshly executed direct time-control optimization. It is a serious feasible-policy and upper-bound calculation on the same original economy, but it is not two

distinct state-space HJB methods and it does not furnish a matched-accuracy advantage for the neural method. A comparative claim requiring monotone PDE, semi-Lagrangian, or controlled Markov-chain baselines at the same certified accuracy remains outside the completed experiment. Likewise, the preserved Riccati-reference study used structured gain outputs; its errors cannot be relabeled as errors of the present 48-by-48 tanh actor and residual-training pipeline.

13.2. Retained comparisons and high-dimensional scope

The historical SOC-MartNet holdouts, method-specific tuning studies, and clipped-feedback comparator are preserved without suppressing unfavorable panels. The clipped-feedback comparator's lower mean cost and the absence of a robust ordering in several higher-dimensional seed panels limit what can be inferred about candidate-generation quality. They do not invalidate the stopped comparison theorem, just as the theorem does not establish empirical superiority.

The 128-dimensional experiment in the retained record certifies a rank-one-coupled Hamiltonian action oracle at frozen jets. Its nonconvex coupling can be reduced to a scalar aggregate search with strongly convex conditional subproblems. This is an action-optimization result. It is not a certified solution of a 128-dimensional dynamic economic model. The actual neural state-domain calculation in this revision concerns the two-state economy in Section 2; no action-dimension result is used to inflate that state dimension.

14. RETAINED R14: WHAT THE CERTIFIED COUNTERFACTUALS IDENTIFY

For any policy, raising k subtracts $(q - p)B(\pi)$, where

$$0 \leq B(\pi) = \mathbb{E} \int_0^\tau e^{-\rho s} \theta_s^2 / 2 \, ds \leq .02 \frac{1 - e^{-\rho}}{\rho}.$$

Consequently V_k is nonincreasing and convex. Monotonicity determines a weak sign without numerical computation, but not a strictly positive or economically well-resolved welfare loss.

Let $[L(k), U(k)]$ be the independently certified central-state value envelope. For $p < q$, a valid loss interval is

$$[\max\{0, L(p) - U(q)\}, \min\{U(p) - L(q), .02C_\rho(1)(q - p)\}], \quad C_\rho(h) = \frac{1 - e^{-\rho h}}{\rho}. \quad (31)$$

A positive lower endpoint establishes strict loss. We report a separate relative-resolution criterion: the width of this interval is no greater than its positive lower endpoint. Neither criterion follows automatically from a .01 bound on policy regret.

The contrast between $2 \rightarrow 8$ and $4 \rightarrow 4.1$ is substantive. The first comparison is strictly positive and relatively resolved by the current intervals. The second has a small absolute upper bound but no certified positive lower bound from interval separation. A policy may therefore be sufficiently accurate for an absolute deployment tolerance while the current value bounds remain too wide for a local comparative-static statement.

The policy library also has a direct economic interpretation. Its selector chooses a stored control vector on each exact price region; within that region no new optimization is required. The supplement lists the selected regions, their widths, consumption ranges, and average preference adjustments. These are properties of feasible chosen policies. They are not claims that the optimal policy itself is piecewise constant in the price, or that policy expenditures equal derivatives of the unknown optimum.

TABLE XI
CERTIFIED WELFARE LOSSES WHEN THE ADJUSTMENT PRICE RISES

p	q	Lower loss	Upper loss	Strict	Relative
0.5	2	0.01266717	0.02940793	Yes	No
2	8	0.02321063	0.04284245	Yes	Yes
4	4.1	0E-8	0.00196053	No	No
4	8	0.00433075	0.02408312	Yes	No
6	8	0E-8	0.01489694	No	No

“Strict” means a strictly positive lower endpoint. “Relative” means interval width no greater than that endpoint. A “No” in the strict column does not reverse the economic monotonicity theorem; it means that strict loss is not separated from zero by these intervals.

TABLE XII
PRICE REGIONS IN WHICH WELFARE LOSSES ARE SEPARATED

Anchor p	Strictly positive loss	Relative width at most one
0.5	(1.093292, 8.0]	(2.617755, 8.0]
2	(2.886123, 8.0]	(6.450558, 8.0]
4	(6.207877, 8.0]	—
6	—	—

Decimal endpoints are descriptive approximations of exact rational thresholds, not proof endpoints. The exact fractions and open-endpoint convention are in the price-audit file. A dash means no such region up to price eight is established by this predicate. The relative criterion here uses the untruncated difference of value intervals and is therefore conservative.

The tolerance .01 is an absolute loss criterion in the utility units of (5). It is not a scale-free welfare measure. The relative and strict-loss diagnostics in (31), rather than the target alone, determine which of the reported economic comparisons are resolved. No empirical calibration or external welfare normalization is imposed to make this declared numerical target appear economically universal.

14.1. A consumption-top-up interpretation

An absolute utility tolerance is not an economic unit. We therefore define a compensation experiment before reporting it. Keep the policy’s preference, wealth, adjustment cost, and stopping paths fixed, and provide an externally financed proportional consumption top-up $\lambda \geq 0$. Only the consumption entering instantaneous utility changes, from c_s to $(1 + \lambda)c_s$. This is an in-kind welfare comparison; it is not a claim that the extra consumption is financeable under the unchanged budget or the original consumption cap.

PROPOSITION 6—A scale-normalized compensation bound: *For a feasible time-control policy with $c \leq 1$, $u \in [1.2, 2.8]$ until stopping, suppose $V - J^\pi \leq \epsilon$ and $\mathbb{E} \int_0^T e^{-.04s} ds \geq A > 0$. If $1 - 1.8\epsilon/A > 0$, a top-up no larger than*

$$\bar{\lambda} = (1 - 1.8\epsilon/A)^{-1/1.8} - 1 \quad (32)$$

is sufficient to compensate the certified policy loss in the specified welfare experiment.

For the fresh central-state price library, the rigorous original stopping calculation supplies

$$A \geq \frac{1 - e^{-.04}}{.04} \left\{ 1 - 2 \exp\left(-\frac{.58^2}{2(.05)^2}\right) \right\}.$$

The resulting uniform top-up bound is 1.033008 percent. It is not assigned to a neural policy. Under a common positive rescaling of the entire welfare criterion, both the loss and the marginal utility gain rescale together, leaving this compensation experiment unchanged. This does not supply an empirical calibration or a wealth-equivalent claim. The exact definition prevents a numerical threshold from being presented as economically meaningful merely because it is slightly below .01.

15. CONCLUSION

A certified neural Bellman method needs an approximation space compatible with the stopping problem, an admissible policy, and complete error inequalities tied to that same policy. The preference-adjustment economy illustrates why these requirements must be checked separately. Its inward upper-wealth boundary creates a strict discontinuity between continuation value and immediate liquidation. The accessibility-preserving architecture and signed facewise theorem remove the resulting trace budget without changing the economic model or restricting the upper comparison's control opportunity.

The new experiments establish distinct advances: complete multi-seed original-economy neural verification with exact trace control; six continuously verified state-space comparison policies; a state-global neural certificate in coupled inventory dynamics up to 128 dimensions; an established-arithmetic recalculation of every decisive historical price node; and fresh end-to-end generation of the full sharp price continuum. A quantitative policy-iteration recurrence states exactly which verified approximation defects imply convergence. A consumption-top-up bound supplies an economic interpretation with an explicit financing convention.

The objective remains an accurate and competitive neural policy/value solver for nonlinear dynamic economies. On the original full-horizon economy, the present neural residual certificates still do not reach .01, and the newly implemented classical candidates do not yet supply a matched-accuracy frontier. The structured inventory success and sharp time-control bounds do not replace those missing nonlinear accuracy results. They supply reproducible constructions, mathematically justified improvements, and separately identified comparison objects through which further numerical advances can be assessed without changing the question being solved.

APPENDIX A: PROOFS OF THE VERIFICATION AND TRANSPORT RESULTS

PROOF OF THEOREM 1: For any admissible control a , stopped Itô's formula gives

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t, z} \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_{a_s} v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

One may first stop the local martingale at a bounded localization sequence. Bounded derivatives and coefficients on the stopped compact domain give integrability, and dominated convergence then yields the displayed identity. For the candidate π , the absolute value of the right side is at most $b + \int_t^T e^{-\rho(s-t)} e(s) ds$. For an arbitrary original control, the residual is at most $e(s) + q(s)$, because its action is covered by the continuous maximum in (8). Hence

$$V(t, z) - v(t, z) \leq b + \int_t^T e^{-\rho(s-t)} \{e(s) + q(s)\} ds.$$

Adding the lower-policy error proves (9). Nonnegativity follows from admissibility of π . If an implemented action belongs to an enclosed neighborhood, the same residual and action-gap checks hold for it by interval inclusion, so the proof is unchanged. No statement about the convergence of the process that generated v or π is used. Q.E.D.

PROOF OF PROPOSITION 1: In the perturbed model, the candidate residual is at least $-e - d$, while the residual of any original admissible action is at most $e + q + d$. Applying the stopped identity separately in that model bounds the lower-policy and upper-value errors by the corresponding integrals. Their sum is $2e + q + 2d$, not an extra improvement-gap penalty on top of those errors. Nonnegative integrands permit replacement of the perturbed discount by $\underline{\rho}$. The stopping-trace error is at most $b + b_{\text{par}}$ if the settlement differs by at most b_{par} . This argument neither equates the two stopping times nor assumes a pathwise coupling of different diffusions. Q.E.D.

APPENDIX B: PROOF OF THE JOINT STATE-PRICE CONSTRUCTION

LEMMA 1—Concavity of the covered policy functional: *For each stored policy satisfying (25), there is a concave function $F_j(z; k)$ of $z \in K$, affine in k , such that $J_k(0, z; \pi_{j,x}) \geq F_j(z; k)$. Its values at the four corners have the independently computed lower endpoints used in the state-price calculation.*

PROOF: Put $r = u - 1$ and $L = -\log c$. The determinant expression in Section 7 has the sign of

$$r^2 L^2 - 2r(r+1)L + 2(r+1) = r^2 L^2 + 2(r+1)(1 - rL) > 0,$$

while $f_{cc} < 0$. Therefore $D^2 f$ is negative definite over the stated region. In the covered Gaussian integral, the preference input is affine in initial preference and consumption is affine in initial wealth. Their image remains in the region just verified. Integrating this concave composition with positive Gaussian and time weights preserves concavity.

The full-horizon preference settlement is the concave quadratic $-.02\{(u - 2 + \int \theta)^2 + .0025\}$. By the budget shift the terminal wealth, and hence its logarithmic settlement, is independent of initial wealth. The deterministic expenditure is independent of the initial state. Subtract a uniform normal-tail and stopping allowance from this covered integral plus its terminal settlement and expenditure. The supplement proves this allowance for all $z \in K$ and $k \in [.5, 8]$. The resulting exact function is F_j , which retains concavity and affine dependence on price. Its coefficient of $-k$ is

$$B_j^{\text{full}} = \frac{1}{2} \int_0^1 e^{-\rho s} \theta_{j,s}^2 ds.$$

At each corner the interval quadrature encloses the exact covered integral and terminal term. Its lower endpoint after the uniform corrections is no larger than F_j there. The finite quadrature rule is not assumed to reproduce the exact functional, and sampled corner values are not used without their enclosure errors. Q.E.D.

PROOF OF PROPOSITION 4: Let the four corners of K be z_v , and write $z = \sum_v \lambda_v(z) z_v$ using the nonnegative bilinear corner weights, which sum to one. Convexity of the verified upper object and concavity of the lower functional imply

$$V_{k_i}(0, z) \leq U_i(z) \leq \sum_v \lambda_v(z) U_i(z_v), \quad J_k(0, z; \pi_{j,x}) \geq F_j(z; k) \geq \sum_v \lambda_v(z) F_j(z_v; k).$$

For k between k_i and k_{i+1} , fixed-policy affinity in price and a common admissible policy set give convexity of V_k . Applying the value chord at the two endpoints yields the chord between the two verified upper corner combinations.

Suppose a policy lower corner endpoint is L_{jv} at k_j and its expenditure belongs to $[B_j^-, B_j^+]$. Its sign-aware lower line is

$$L_{jv} - (k - k_j)B_j^+ \quad (k \geq k_j), \quad L_{jv} - (k - k_j)B_j^- \quad (k \leq k_j).$$

Each price cell between consecutive nodes has a fixed sign for every policy. The upper chord minus this lower line is therefore an affine function $g_{jv}(k)$. For every state, the regret of policy j is at most $\sum_v \lambda_v(z) g_{jv}(k) \leq \max_v g_{jv}(k)$. Selecting a policy minimizing this last bound proves the uniform state guarantee

$$R(k) = \min_j \max_v g_{jv}(k).$$

Take all pairwise intersections of the finitely many affine functions and the price-cell endpoints. Between successive such points, their ordering does not change. The maximum for each policy and then the minimum across policies select a single affine function on that interval. Its maximum occurs at an endpoint. Exact rational evaluation at this finite set thus bounds the supremum over all real prices. A corner line is discarded only when another corner line from the same policy dominates it at both price-cell endpoints, which proves domination on the entire cell. The program records those reductions and checks the remaining intersections without a sampled-grid extrapolation. *Q.E.D.*

PROOF OF PROPOSITION 5: The independent primal and dual constructions in the supplement supply, respectively, valid original-policy lower endpoints and upper functions (24). Exact action and terminal-budget checks hold for every central policy and every shifted policy at the extremes of the initial-state rectangle. Affinity of the shift extends these feasibility checks to its interior. Lemma 1 supplies the concave lower functional with a common tail and stopping allowance. All quantities in Proposition 4 are therefore present.

The execution uses the 18 prices

$$.5, .875, 1.25, 1.625, 2, 2.5, 3, 3.5, 4, 4.25, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8.$$

The stored state–price record checks 211 candidate breakpoints and returns the exact fraction displayed following (27); upward decimal conversion gives (26). The central-state calculation uses the exact stopped-policy expenditure intervals and checks 187 candidate breakpoints with a separate all-intersections implementation. Its exact worst-case fraction is

$$\frac{1600901779824582944359922627717}{160587876576305697877131733762048},$$

whose upward display is 0.009969007711. These are computer-assisted conclusions conditional on the explicitly stated arithmetic model and analytic enclosure lemmas, not consequences of agreement between two floating-point approximations. *Q.E.D.*

APPENDIX C: ECONOMIC RESOLUTION AND THE COMMON-ENDPOINT STRESS BOUND

For $p < q$, every policy loses $(q - p)B(\pi) \geq 0$, and $B(\pi) \leq .02C_\rho(1)$. Taking suprema preserves monotonicity and gives $0 \leq V_p - V_q \leq .02C_\rho(1)(q - p)$. Combining this fact with the independently verified intervals at p and q gives (31). On any segment where the upper and lower

envelope pieces are affine, both strict separation and the conservative relative-width inequality reduce to affine inequalities in q . Their roots, included with the exact envelope breakpoints, produce the resolution regions without a grid test.

Finally, if every nodal upper endpoint is enlarged by δ and every nodal lower endpoint is decreased by δ , every upper chord is enlarged by δ and every lower policy line is decreased by δ . Their maximum/minimum envelope gap increases by at most 2δ . This proves the recorded endpoint stress test and also its scope: it tests sensitivity to valid endpoint inflation, not the validity of upstream node proofs.

APPENDIX D: PROOFS FOR STOPPING ACCESSIBILITY AND CERTIFIED ITERATION

PROOF OF PROPOSITION 2: Fix $x_0 \in [1.5, 2)$ and use $c = .8$, $\theta = p = 0$. Wealth decreases, stays below two, and satisfies $x_s \geq x_0 - .8s \geq .7$ for $s \leq 1$. Thus it does not hit either wealth face. Preferences equal $2 + .05W_s$. On the event $H = \{\sup_{s \leq 1} |.05W_s| \leq .2\}$ there is no preference exit, $u \in [1.8, 2.2]$, and running utility is at least -1.5 . To see the last inequality, set $r = u - 1 \in [.8, 1.2]$. The function $-(5/4)^r/r$ is increasing on this interval since $r \log(5/4) < 1$. Its value at $.8$ exceeds -1.5 , using $(5/4)^4 < (6/5)^5$. Settlement on H is at least $-.05$, because $-.02(.2)^2 + .1 \log(.7) > -.05$. Discounting negative terms cannot worsen these lower bounds. The pathwise discounted payoff on H is therefore at least -1.55 .

Until the original stopping time, $u \in [1.2, 2.8]$. The same elementary monotonicity bounds running utility by -6 from below, and $G \geq -.1$. Even on H^c , the stopped payoff is at least $-6 - 8 - .1 = -14.1$. The Brownian maximal inequality gives $\Pr(H^c) \leq 2e^{-8}$. Hence

$$J_2(0, 2, x_0; .8, 0, 0) \geq -1.55 - 12.55(2e^{-8}).$$

Subtracting $g(0, 2, 2) = .1 \log 2 - 8$ proves (12), uniformly as $x_0 \uparrow 2$. Directed MPFR evaluation encloses its right-hand side in the interval recorded in the replication; its lower endpoint exceeds 6.37.

Finally, a smooth witness equal to g on all stopping faces and with $\sup_a \mathcal{R}_a v \leq e_+(s)$ gives $V(0, z) \leq v(0, z) + \int_0^1 e^{-.04s} e_+(s) ds$ by the stopped Itô identity. Take $z = (2, x)$, let $x \uparrow 2$, and use continuity of v and the established strict gap. The integrated uniform positive residual must exceed 6.37. A sequence with vanishing such residuals is excluded, without making any assertion that the economic control problem lacks a solution. *Q.E.D.*

PROOF OF THEOREM 2: Discounted Itô's formula, localization of the stochastic integral, and the bounded stopped-domain hypotheses give, for every admissible control a ,

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t,z}^a \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_a v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

The positive residual bound and the facewise upper discrepancy bounds imply

$$V - v \leq \int_t^T e^{-\rho(s-t)} e_+(s) ds + \sum_i \beta_i^+ \omega_i^*.$$

Apply the same identity to π with the opposite sign to obtain

$$v - J^\pi \leq \int_t^T e^{-\rho(s-t)} e_-(s) ds + \sum_i \beta_i^- \omega_i^\pi.$$

Adding proves the assertion. No common stopping time across two distinct controlled paths is assumed. For the proposed smooth hitting-probability bound w_i , optional stopping yields $\mathbb{E}[e^{-\rho(\tau-t)}w_i(\tau, Z_\tau)] \leq w_i(t, z)$; domination of the indicator on the stopping boundary then gives the stated probability bound. *Q.E.D.*

PROOF OF PROPOSITION 3: Write $Y = 2 - x > 0$ and $p = Yq$, where $q = (.15 + .65 \tanh N_p)/1.5$ is bounded. Until another exit,

$$dY = (c - .04)ds + Y\{.02 - .06xq\}ds - .2xqY dZ_s^x.$$

The bounded coefficients define a strictly positive stochastic exponential Φ . Variation of constants gives

$$Y_s = \Phi_s \left(Y_t + \int_t^s \Phi_r^{-1}(c_r - .04)dr \right) > 0,$$

since $c \geq .05$. The upper wealth face is therefore inaccessible from the interior. On the remaining state faces $B = 0$; at terminal time $1 - t = 0$. On the upper wealth face $B \geq 0$ and softplus is positive, so $v - g \geq 0$. Thus every upper-comparison trace discrepancy is zero, and the only potentially positive policy discrepancy has zero hitting probability. The same calculation applies to admissible bounded clipped head perturbations before the distance multiplication. *Q.E.D.*

PROOF OF THEOREM 3: Let $d_n = \epsilon_n/(1 - q)$. The resolvent of $I - qP_{\pi_n}$ is nonnegative and has row sums at most $(1 - q)^{-1}$, so $\|v_n - V^{\pi_n}\|_\infty \leq d_n$. Put $a_n = 2qd_n + \eta_n$. The contraction inequalities and approximate greedy inequality give

$$TV^{\pi_n} - T_{\pi_{n+1}}V^{\pi_n} \leq a_n \mathbf{1}.$$

Since $TV^{\pi_n} \geq V^{\pi_n}$, the same nonnegative resolvent gives $V^{\pi_{n+1}} \geq V^{\pi_n} - a_n \mathbf{1}/(1 - q)$. Therefore

$$\begin{aligned} V^* - V^{\pi_{n+1}} &= TV^* - T_{\pi_{n+1}}V^{\pi_{n+1}} \\ &\leq qE_n \mathbf{1} + a_n \mathbf{1} + qP_{\pi_{n+1}}(V^{\pi_n} - V^{\pi_{n+1}}) \\ &\leq \{qE_n + a_n/(1 - q)\} \mathbf{1}. \end{aligned}$$

Taking the supremum norm proves (17). The rate follows by summing the resulting geometric convolution. The comparison with a continuous-time problem is the triangle inequality after separately bounding the two transfer terms; it does not follow from the finite Markov contraction alone. *Q.E.D.*

APPENDIX E: PROOF OF THE STATE-GLOBAL INVENTORY CERTIFICATE

PROOF OF THEOREM 4: The scalar backward Riccati solutions exist on the finite horizon, remain at least .5, and are bounded above. Indeed, in remaining time their drift is $2a_i p - p^2 + q_i$, positive at .5 and negative above its positive equilibrium. The neural gains also remain at least .5 by construction. Let $d_i = \hat{p}_i - p_i^*$ and let R_i be the neural Riccati residual. Then

$$d'_i = (\hat{p}_i + p_i^* - 2a_i)d_i + R_i, \quad d_i(1) = 0.$$

The coefficient is at least $\kappa_i = 1 - 2a_i$. Integrating backward therefore yields $|d_i(t)| \leq \varepsilon_i(1 - e^{-\kappa_i(1-t)})/\kappa_i \leq D$.

The quadratic optimal value has matrix $P^* = p_\perp^*(I - M) + p_m^*M$ and a scalar noise term. Completing the square in its generator and applying Itô's formula gives the exact identity

$$\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x) = \mathbb{E} \int_0^1 |\{\hat{P}(s) - P^*(s)\}X_s|^2 ds.$$

The two projectors are orthogonal, so the integrand is bounded by $D^2|X_s|^2$. Both eigenvalues of $A - \hat{P}$ are at most $-.4$. Itô's formula for $|X_s|^2$ then gives

$$\frac{d}{ds} \mathbb{E}|X_s|^2 \leq -.8\mathbb{E}|X_s|^2 + .01d.$$

Gronwall's inequality and $|x|^2/d \leq 1$ imply $d^{-1}\mathbb{E}|X_s|^2 \leq e^{-.8s} + (.01/.8)(1 - e^{-.8s})$. Integration proves (30). The bounded deterministic gains give finite second moments and justify localization and removal of the stopping sequence in both Itô identities. No bounded-state truncation or simulated tail probability is used. *Q.E.D.*

APPENDIX F: PROOF OF THE COMPENSATION BOUND

PROOF OF PROPOSITION 6: Holding the specified paths fixed, the derivative of consumption utility with respect to the top-up is

$$\frac{\partial}{\partial \lambda} \frac{\{(1 + \lambda)c\}^{1-u}}{1 - u} = c^{1-u}(1 + \lambda)^{-u} \geq (1 + \lambda)^{-2.8}.$$

The inequality uses $c \leq 1$, $u > 1$, $u \leq 2.8$, and $\lambda \geq 0$. Integrating from zero to λ , then over the discounted alive-time measure, bounds the welfare gain below by $A\{1 - (1 + \lambda)^{-1.8}\}/1.8$. Solving for a gain of ϵ proves (32). For the time-control library, wealth feasibility excludes a wealth exit, and the retained drift bounds and Brownian maximal inequality give the displayed preference-exit probability. Surviving to terminal time alone yields the stated lower bound on discounted alive time. This argument requires no extrapolation of the certified optimal value to an altered wealth state. *Q.E.D.*

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